

MKTG 638 Database Marketing Project Report

Title: Drivers of Product Success and Revenue on Amazon

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Main Content

Purpose of Project

The purpose of this project is to conduct a comprehensive analysis of e-commerce sales performance and identify the key factors influencing revenue generation. The specific objectives are as follows:

- **Identify Top-Performing Products:** Determine which products generate the highest sales and revenue, and evaluate their contribution to overall business performance.
- **Segment Products by Performance:** Classify products into distinct performance tiers based on sales metrics, enabling clearer strategic decision-making.
- **Analyze Pricing and Rating Patterns:** Examine the relationship between product pricing, customer ratings, and sales outcomes to identify meaningful trends and correlations.
- **Evaluate Category-Level Performance:** Assess the performance of different product categories to determine which segments drive the most value.
- **Conduct Advanced Revenue Driver Analysis:** Perform in-depth analytical modeling to identify and quantify the key factors that most significantly influence revenue generation.

Research Questions

Research Question A: Do high-quantity orders in North America represent the top-performing transactions in terms of units sold?

Research Question B: Do “Excellent” rated products generate significantly higher revenue than “Good” or “Poor” rated products?

Research Question C: What is the overall purchasing behavior across all Amazon orders, and is there a measurable relationship between `discount_percent` and `quantity_sold`?

Research Question D: Which product categories demonstrate strong revenue efficiency by generating an average revenue per order greater than \$300?

Research Question E: Within each `customer_region`, how do `payment_methods` rank based on the number of orders, and which payment method is most preferred in each region?

Dataset Description

This dataset used is simulated Amazon-style e-commerce sales data designed for use in data analysis, visualization, and machine learning applications. It reflects realistic time-series sales trends, including variations in pricing, discounts, customer geographic distribution, and product ratings across multiple product categories.

Link to dataset:

<https://www.kaggle.com/datasets/aliiihussain/amazon-sales-dataset>

Column	Definition
order_id	A unique identifier assigned to each customer transaction. It distinguishes one order from another and allows tracking of individual purchases.
order_date	The date on which the order was placed. This variable is used to analyze sales trends over time (daily, monthly, seasonal patterns).
product_id	A unique identifier for each product. It allows grouping and aggregation of sales performance at the product level.
product_category	The classification or category to which a product belongs (e.g., Electronics, Clothing, Home & Kitchen). Used for category-level performance analysis.
price	The original listed price of the product before any discount is applied.
discount_percent	The percentage discount applied to the product at the time of purchase. A value of 0 indicates no discount.
discounted_price	The final price paid per unit after applying the discount. Typically calculated as: $\text{price} \times (1 - \text{discount_percent})$
quantity_sold	The number of units purchased in a single order transaction.
total_revenue	The total revenue generated from the order line. Typically calculated as: $\text{discounted_price} \times \text{quantity_sold}$
customer_region	The geographic region where the customer is located (e.g., North, South, East, West). Used for regional sales comparison.
payment_method	The method used to complete the transaction (e.g., Credit Card, Debit Card, PayPal, Cash on Delivery). Helps analyze payment behavior patterns.
rating	The average customer rating of the product, typically measured on a 1–5 scale. Indicates customer satisfaction.
review_count	The total number of customer reviews submitted for the product. Often used as a proxy for product popularity and customer engagement.

SQL Output: A

	order_id bigint	order_date text	product_id bigint	product_category text	quantity_sold integer	total_revenue numeric (14,2)
1	27833	2022-03-04	3821	Sports	5	2499.55
2	18858	2022-06-20	1570	Sports	5	2487.30
3	20016	2022-02-02	3667	Electronics	5	2479.15
4	22400	2022-12-02	1024	Fashion	5	2472.30
5	36495	2022-12-23	1383	Beauty	5	2462.30
6	32159	2022-09-02	3994	Electronics	5	2454.50
7	44939	2023-04-09	3452	Books	5	2451.05
8	43794	2022-03-01	1580	Books	5	2434.70
9	35373	2022-03-07	1404	Books	5	2428.05
10	1077	2023-06-01	2074	Fashion	5	2419.35

	product_id bigint	product_category text	units_sold_in_top10 bigint	revenue_in_top10 numeric	rows_in_top10 bigint
1	3821	Sports	5	2499.55	1
2	1570	Sports	5	2487.30	1
3	3667	Electronics	5	2479.15	1
4	1024	Fashion	5	2472.30	1
5	1383	Beauty	5	2462.30	1
6	3994	Electronics	5	2454.50	1
7	3452	Books	5	2451.05	1
8	1580	Books	5	2434.70	1
9	1404	Books	5	2428.05	1
10	2074	Fashion	5	2419.35	1

	product_id bigint	product_revenue numeric	top10_total_revenue numeric	pct_of_top10_revenue numeric
1	3821	2499.55	24588.25	10.17

Interpretation of Output A

The analysis of the top 10 North American transactions ranked by quantity_sold shows that all selected orders reflect the highest unit performance in the dataset, with each transaction recording 5 units sold. This confirms that high-quantity orders in North America do represent the top-performing transactions in terms of units sold. Because the results were filtered by region and ordered in descending quantity, the output captures the strongest unit-level performers within the specified market.

In terms of revenue performance, the total revenue generated by these top 10 transactions amounts to approximately \$24,588.25. The highest-revenue product within this group contributes about \$2,499.55, accounting for roughly 10.17% of the total revenue. The remaining revenue is distributed across several other products spanning multiple categories, including Sports, Electronics, Fashion, Books, and Beauty. The revenue values across these products fall within a relatively narrow range, indicating similar revenue contributions among the top-performing unit transactions.

The limited revenue concentration suggests a balanced revenue structure within high-quantity sales. No single product or category overwhelmingly dominates revenue among the top-performing transactions. From a managerial perspective, this indicates diversified product demand in the North American market. While high-quantity orders clearly define top performance in terms of units sold, the revenue generated within those transactions reflects broad marketplace participation rather than dependence on a single dominant product or category.

SQL Output: B

	rating_category character varying (20)	total_orders bigint	avg_revenue numeric	total_revenue numeric
1	Excellent	6798	661.40	4496210.88
2	Good	12622	659.58	8325204.52
3	Poor	30580	655.50	20045158.34

Interpretation of Output B

The analysis includes 50,000 total orders generating approximately \$32.87 million in revenue. Order volume is heavily concentrated in the “Poor” category, which accounts for 30,580 orders (61.2% of total orders). The “Good” category represents 12,622 orders (25.2%), while “Excellent” accounts for 6,798 orders (13.6%). Revenue distribution closely mirrors this pattern: “Poor” contributes about \$20.05 million (61.0%), “Good” generates \$8.33 million (25.3%), and “Excellent” brings in \$4.50 million (13.7%). This alignment between order share and revenue share indicates that total revenue differences are primarily driven by volume rather than spending differences across rating groups.

When examining average revenue per order—the most meaningful metric for comparing spending behavior—the values are nearly identical across categories: \$661.40 for “Excellent,” \$659.58 for “Good,” and \$655.50 for “Poor,” representing less than a 1% difference between the highest and lowest groups. Although “Excellent” rated products generate slightly higher revenue on average, the difference is minimal and does not suggest a strong relationship between rating category and revenue performance. Overall, product ratings appear to have limited impact on revenue per order, with overall revenue outcomes largely influenced by the distribution of orders across categories.

SQL Output: C

	total_orders bigint	min_quantity integer	max_quantity integer	avg_quantity numeric	quantity_std_dev numeric	discount_quantity_correlation double precision
1	50000	1	5	2.9994000000000000	1.4154008998139516	0.0013989282759390089

Interpretation of Output C

The analysis evaluates purchasing behavior across Amazon orders by examining the distribution of units purchased per transaction and assessing whether discount levels influence order volume. The results show that customers purchase between 1 and 5 units per order, indicating that transaction sizes remain within a narrow band and that the dataset does not reflect extreme bulk-buying patterns. The average quantity sold per order is close to three units, suggesting that most customer purchases are moderate in size and relatively consistent across transactions. The standard deviation indicates some variation in order quantities, but not enough to suggest highly uneven purchasing behavior or large outliers; overall, order volume appears stable and predictable.

To understand the impact of promotions, the correlation between `discount_percent` and `quantity_sold` was calculated. The correlation value is extremely close to zero, meaning there is no meaningful linear relationship between the level of discount offered and the number of units customers purchase per order. In practical terms, this suggests that increasing discounts does not significantly encourage customers to buy more units within a transaction. From a managerial perspective, this finding implies that blanket discounting strategies may be ineffective if the objective is to increase purchase quantities. Instead, Amazon should consider more targeted approaches—such as product bundles, category-specific promotions, or region-focused campaigns—since customer purchase volume appears to be driven more by underlying demand and purchase intent than by discount depth alone.

SQL Output: D

	product_category text	avg_revenue_per_order numeric
1	Home & Kitchen	662.7673225962702834
2	Books	658.6841635643088747
3	Electronics	657.5233209134615385
4	Beauty	655.7147040756054341
5	Fashion	655.1253245666467424
6	Sports	654.2330090744101633

Interpretation of Output D

The analysis shows that all product categories generate an average revenue per order greater than \$300, indicating strong overall revenue efficiency across the dataset. In fact, the average revenue per order for each category falls within a narrow range of approximately \$654 to \$663. Among the categories, Home & Kitchen demonstrates the highest average revenue per order at approximately \$662.77, followed closely by Books and Electronics. The remaining categories—Beauty, Fashion, and Sports—also maintain similarly high averages, all exceeding \$650 per order. This suggests that customers tend to make relatively high-value purchases regardless of product category.

The minimal variation between categories indicates a balanced revenue structure, with no single category significantly outperforming or underperforming the others in terms of average order value. Because all categories exceed the defined \$300 benchmark by a substantial margin, the company demonstrates consistent revenue efficiency across its product portfolio. From a managerial perspective, this balanced performance suggests diversified revenue streams and reduces dependency on any one category, supporting stable financial performance and strategic flexibility in marketing and pricing decisions.

SQL Output: E

	customer_region text	payment_method text	order_count bigint	payment_rank bigint
1	Asia	UPI	2606	1
2	Asia	Wallet	2496	2
3	Asia	Cash on Delivery	2487	3
4	Asia	Credit Card	2486	4
5	Asia	Debit Card	2451	5
6	Europe	Debit Card	2551	1
7	Europe	Wallet	2544	2
8	Europe	UPI	2478	3
9	Europe	Credit Card	2468	4
10	Europe	Cash on Delivery	2411	5
11	Middle East	Cash on Delivery	2544	1
12	Middle East	Wallet	2526	2
13	Middle East	Credit Card	2493	3
14	Middle East	Debit Card	2486	4
15	Middle East	UPI	2456	5
16	North America	Wallet	2540	1
17	North America	UPI	2538	2
18	North America	Debit Card	2493	3
19	North America	Cash on Delivery	2485	4
20	North America	Credit Card	2461	5

Interpretation of Output E

The analysis evaluates customer payment preferences within each region by ranking payment methods based on total order count. The results indicate that payment preferences vary slightly across regions, although overall order distribution across payment methods remains relatively balanced.

In **Asia**, UPI ranks as the most preferred payment method, followed closely by Wallet and Cash on Delivery, while Debit Card records the lowest order count within the region. This suggests strong adoption of digital payment infrastructure, particularly UPI-based systems.

In **Europe**, Debit Card emerges as the leading payment method, with Wallet and UPI following closely behind. Cash on Delivery ranks lowest, indicating a stronger preference for electronic and card-based transactions over traditional payment methods.

In the **Middle East**, Cash on Delivery holds the top position, with Wallet and Credit Card closely competing for subsequent ranks. UPI records the lowest preference in this region, reflecting regional differences in digital payment penetration.

In **North America**, Wallet is the most preferred payment method, narrowly surpassing UPI. Credit Card ranks lowest in this region, though the difference in order counts across payment types remains marginal.

Across all regions, the variation in order counts between the highest- and lowest-ranked payment methods is relatively small. This indicates diversified payment behavior rather than heavy dependence on a single transaction method within any specific market.

From a managerial perspective, these findings suggest that while regional preferences exist, Amazon benefits from a well-balanced payment ecosystem. Strategic focus should involve strengthening regionally dominant payment methods, such as UPI in Asia or Cash on Delivery in the Middle East, while continuing to support multiple payment options to maintain flexibility and customer convenience across markets.

Recommendations

Diversified Product Promotion

From a marketing perspective, the findings suggest that demand in North America is not driven by a single dominant product, but rather by a diverse mix of categories that consistently generate high unit sales. Because revenue among the top-performing transactions is relatively balanced, marketing efforts should focus on promoting a range of high-demand products rather than concentrating resources on one flagship item. Highlighting these strong-performing products in targeted campaigns, seasonal promotions, and personalized recommendations can help maintain momentum in unit sales. Additionally, bundling complementary products across categories may increase overall order value while reinforcing cross-category purchasing behavior. Since North America shows strong high-quantity performance, continued regional

targeting and inventory alignment will be essential to sustain growth and prevent stock shortages during peak demand periods.

Performance-based Segmentation

Since the difference in average revenue between “Excellent”, “Good”, and “Poor” products is minimal, it does not strongly justify heavy marketing investment based solely on rating category. A small gap suggests that rating alone is not a strong indicator of revenue and heavy marketing investments in each category would likely produce a small incremental return. Instead, the rating category should be a supporting factor to other comprehensive performance-indicators such as total revenue, profit margin, sales volume, or growth rate, allowing informed marketing decisions to be made.

Promotional Strategy

Since discount levels do not significantly influence purchase quantity, Amazon should avoid relying on blanket discounting strategies to drive higher order volumes. Instead, promotional efforts should be more targeted and data-driven, focusing on specific products or customer segments where price sensitivity is higher. Bundled offers and value-added packages may be more effective in increasing units per transaction than simply increasing discount percentages. Regional pricing experiments can help identify localized responsiveness before scaling discount strategies. Additionally, marketing efforts should emphasize product value and differentiation rather than price reductions alone. This approach will help protect margins while maintaining steady sales performance. A shift toward strategic promotion rather than aggressive discounting will support long-term profitability and sustainable growth.

Cross-targeting

Based on the results, all product categories demonstrate strong and consistent revenue efficiency, with average revenue per order exceeding \$650, which is well above the defined \$300 benchmark. Since performance is relatively balanced across categories, the company does not appear to rely heavily on a single product segment for revenue generation, indicating a diversified and stable sales structure. Rather than making major structural changes, management should focus on strategies that incrementally increase average order value, such as upselling premium products, offering bundled promotions, or implementing targeted cross-selling campaigns within each category. Additionally, slightly higher-performing categories like Home & Kitchen may receive modest additional marketing support to further capitalize on their marginal advantage. Overall, the findings suggest strong revenue stability, and future efforts should prioritize revenue optimization and value enhancement rather than corrective action.

Payment Method Strategy

Amazon should strengthen and optimize regionally dominant payment methods to enhance customer convenience and checkout efficiency. In Asia, expanding UPI partnerships and promotional incentives may further reinforce digital payment adoption. In the Middle East, maintaining strong Cash on Delivery infrastructure remains essential to support local preferences. Europe and North America should continue enhancing digital wallet and

card-based payment integrations to improve user experience. Offering localized payment promotions can further align with regional behavior. At the same time, maintaining a diversified payment ecosystem ensures operational flexibility and risk mitigation. A region-specific payment optimization strategy will improve customer satisfaction while supporting stable transaction growth.

Appendix

SQL Coding: A

SELECT
FROM
WHERE
ORDER BY
LIMIT

SQL CODE:

```
SELECT  
  Order_id,  
  Order_date,  
  Product_id,  
  Product_category,  
  Quantity_sold,  
  Total_revenue  
FROM sales  
WHERE customer_region = 'North America'  
ORDER BY quantity_sold DESC,  
total_revenue DESC  
LIMIT 10;
```

```
21 SELECT  
22   order_id,  
23   order_date,  
24   product_id,  
25   product_category,  
26   quantity_sold,  
27   total_revenue  
28 FROM sales  
29 WHERE customer_region = 'North America'  
30 ORDER BY quantity_sold DESC, total_revenue DESC  
31 LIMIT 10;  
  
32 WITH top10_na AS (  
33   SELECT  
34     order_id,  
35     product_id,  
36     product_category,  
37     quantity_sold,  
38     total_revenue  
39   FROM sales  
40   WHERE customer_region = 'North America'  
41   ORDER BY quantity_sold DESC, total_revenue DESC  
42   LIMIT 10  
43 )  
44 SELECT  
45   product_id,  
46   product_category,  
47   SUM(quantity_sold) AS units_sold_in_top10,  
48   SUM(total_revenue) AS revenue_in_top10,  
49   COUNT(*) AS rows_in_top10  
50 FROM top10_na  
51 GROUP BY product_id, product_category  
52 ORDER BY units_sold_in_top10 DESC, revenue_in_top10 DESC;  
  
53 WITH top10_na AS (  
54   SELECT product_id, total_revenue  
55   FROM sales  
56   WHERE customer_region = 'North America'  
57   ORDER BY quantity_sold DESC, total_revenue DESC  
58   LIMIT 10  
59 ),  
60 product_rev AS (  
61   SELECT product_id, SUM(total_revenue) AS product_revenue  
62   FROM top10_na  
63   GROUP BY product_id  
64 ),  
65 totals AS (  
66   SELECT SUM(total_revenue) AS top10_total_revenue  
67   FROM top10_na  
68 )  
69 SELECT  
70   pr.product_id,  
71   pr.product_revenue,  
72   t.top10_total_revenue,  
73   ROUND(100.0 * pr.product_revenue / t.top10_total_revenue, 2) AS pct_of_top10_revenue  
74 FROM product_rev pr  
75 CROSS JOIN totals t  
76 ORDER BY pr.product_revenue DESC  
77 LIMIT 1;
```

Step 1: Filter for North America transactions.

- Used a WHERE clause to isolate transactions where customer_region = 'North America'.
- This ensures the analysis focuses only on the target geographic market.

Step 2: Rank transactions by units sold.

- Used ORDER BY quantity_sold DESC to sort transactions from highest to lowest unit volume.
- This identifies which transactions represent the strongest performance in terms of units sold.

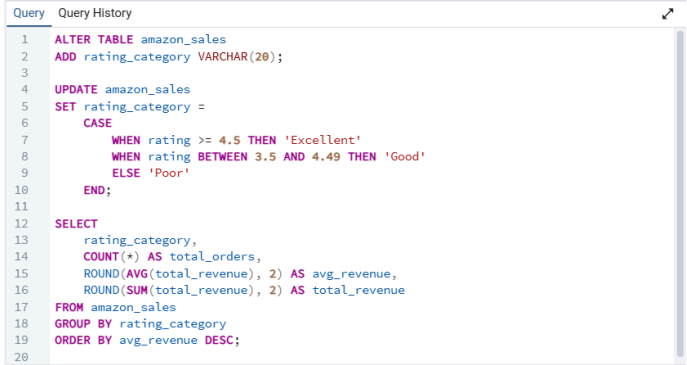
Step 3: Limit results to top-performing transactions.

- Applied LIMIT 10 to retrieve only the top 10 highest-quantity transactions.
- This allows direct evaluation of the strongest-performing orders within North America.

Step 4: Evaluate product and revenue concentration.

	- Grouped results by product_id and product_category. - SUM(quantity_sold) identifies best-selling products within the top 10. - SUM(total_revenue) measures total revenue contribution per product. - Revenue share calculation determines whether performance is concentrated in a single product or distributed across multiple categories.
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SQL Coding: B

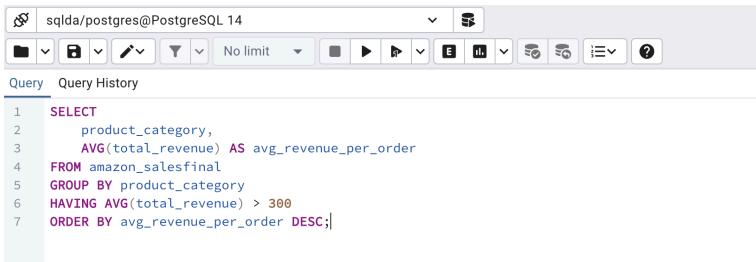
CASE WHEN condition1 THEN value 1 WHEN condition 2 THEN value 2 WHEN condition 3 THEN value 3 SQL CODE: ALTER TABLE amazon_sales ADD rating_category VARCHAR(20); UPDATE amazon_sales SET rating_category = CASE WHEN rating >= 4.5 THEN 'Excellent' WHEN rating BETWEEN 3.5 AND 4.49 THEN 'Good' ELSE 'Poor' END; SELECT rating_category, COUNT(*) AS total_orders, ROUND(AVG(total_revenue), 2) AS avg_revenue, ROUND(SUM(total_revenue), 2) AS total_revenue FROM amazon_sales GROUP BY rating_category ORDER BY avg_revenue DESC;	 Step 1: Create a new column titled “rating_category”. - Added a new column, rating_category. - Column allows numerical ratings to be compared to categorical labels. Step 2: UPDATE with CASE. - Assigned each product a category based on rating. - Created comparable groups which were easy to interpret. Step 3: SELECT Query. - Rating_category: categories being compared. - COUNT(*) AS total_orders: number of orders in each category. - AVG(total_revenue): averages revenue for each order, for each category. - SUM(total_revenue): shows total revenue for each category.
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SQL Coding: C

COUNT (columnX) MIN (columnX) MAX (columnX) AVG (columnX) STDDEV(columnX) CORR(columnX,columnY) SQL CODE:	<pre>SELECT COUNT(quantity_sold) AS total_orders, MIN(quantity_sold) AS min_quantity, MAX(quantity_sold) AS max_quantity, AVG(quantity_sold) AS avg_quantity, STDDEV(quantity_sold) AS quantity_std_dev, CORR(discount_percent, quantity_sold) AS discount_quantity_correlation FROM amazon_sales;</pre> Step 1: Evaluate purchasing behavior
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<pre>SELECT COUNT(quantity_sold) AS total_orders, MIN(quantity_sold) AS min_quantity, MAX(quantity_sold) AS max_quantity, AVG(quantity_sold) AS avg_quantity, STDDEV(quantity_sold) AS quantity_std_dev, CORR(discount_percent, quantity_sold) AS discount_quantity_correlation FROM amazon_sales;</pre>	• Applied aggregate functions on quantity_sold to summarize order patterns. • COUNT() measured total transactions. • MIN() and MAX() identified order size range. • AVG() determined typical units per order. • STDDEV() assessed variability in purchase quantities. • This provides a statistical overview of customer buying consistency. Step 2: Assess discount impact • Used CORR(discount_percent, quantity_sold) to measure the linear relationship between discounts and order volume. • This determines whether higher discounts are associated with increased purchase quantities. Step 3: Interpret strategic implications • Compared variability and correlation results to evaluate whether purchasing behavior is discount-driven or structurally consistent. • This connects statistical findings to promotional strategy decisions.
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SQL Coding: D

SELECT FROM GROUP BY HAVING ORDER BY SQL CODE: SELECT product_category, AVG(total_revenue) AS avg_revenue_per_order FROM amazon_salesfinal GROUP BY product_category HAVING AVG(total_revenue) > 300 ORDER BY avg_revenue_per_order DESC;	 Step 1: Aggregate revenue by product category • Used AVG(total_revenue) to calculate average revenue per order. • Grouped results using GROUP BY product_category. Step 2: Apply performance threshold • Used HAVING AVG(total_revenue) > 300. • Filtered categories to identify strong revenue efficiency. Step 3: Rank results • Applied ORDER BY avg_revenue_per_order DESC. • Ranked categories from highest to lowest average revenue.
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SQL Coding: E

<pre>SELECT RANK () OVER (PARTITION BY ORDER BY) AS SQL CODE: SELECT customer_region, payment_method, order_count, RANK() OVER (PARTITION BY customer_region ORDER BY order_count DESC) AS payment_rank FROM (SELECT customer_region, payment_method, COUNT(order_id) AS order_count FROM amazon_sales GROUP BY customer_region, payment_method) sub;</pre>	<pre>SELECT customer_region, payment_method, order_count, RANK() OVER (PARTITION BY customer_region ORDER BY order_count DESC) AS payment_rank FROM (SELECT customer_region, payment_method, COUNT(order_id) AS order_count FROM amazon_sales GROUP BY customer_region, payment_method) sub;</pre> Step 1: Aggregate order volume by region and payment method • Grouped data using GROUP BY customer_region, payment_method. • Used COUNT(order_id) to calculate total orders per payment method within each region. • Summarizes payment distribution across regions. Step 2: Rank payment methods within each region • Applied RANK() with PARTITION BY customer_region. • Ordered by order_count DESC to rank payment methods from highest to lowest usage. • Identifies the most preferred payment method in each region. Step 3: Interpret regional payment behavior • Compared rankings across regions. • Evaluated dominance and variation in payment preferences. • Supports region-specific payment strategy decisions.
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